



PERTH MODERN SCHOOL
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Independent Public School

Course Methods

Year 11

Student name: _____

Teacher name: _____

Date: 27/07/20

Task type: Response

Time allowed for this task: 30 mins

Number of questions: 5

Materials required: NO CALCULATORS ALLOWED
ONE A4 PAGE BOTH SIDES OF NOTES ALLOWED
FORMULA SHEET PROVIDED

Standard items: Pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: Drawing instruments, templates and formula sheet

Marks available: 30 marks

Task weighting: 10 %

Formula sheet provided: Yes

Note: All part questions worth more than 2 marks require working to obtain full marks.

Question 1 (1.3.2)**(2, 2 = 4 marks)**

Evaluate and express your answer in whole numbers.

i) $6!$

ii) $\binom{10}{6}$

Question 2 (1.3.1)**(2, 3 = 5 marks)**

- a) Expand $(1 - x)^4$ in ascending powers of x . Express your answer as whole numbers.
- b) Show how you would use your answer in (a) to calculate the value of 0.99^4 . State this value correct to 4 decimal places.

Question 3 (1.3.2)**(1, 1, 1, 2, 2 = 7 marks)**

The Australian Chess team of 9 people is to be selected from 10 from West Australia, 8 from NSW and 5 from Victoria. Write mathematical expressions for the number of different ways the team can be selected if:

- a) There are no restrictions

- b) All three states are equally represented.

- c) There are no Victorians

- d) The NSW representatives are in the majority

- e) The WA twins Bill and Ben, cannot be in the same team.

Question 4 (1.2.7)**(1, 1, 1, 1, 2, 2 = 8 marks)**

The diagram shows a unit circle with centre O . A is a point on the unit circle with co-ordinates (p, q) . The ray OA is inclined at an angle of 25° to the positive x -axis as shown. Use the unit circle to find in terms of p and/or q :

a) $\cos -25^\circ$

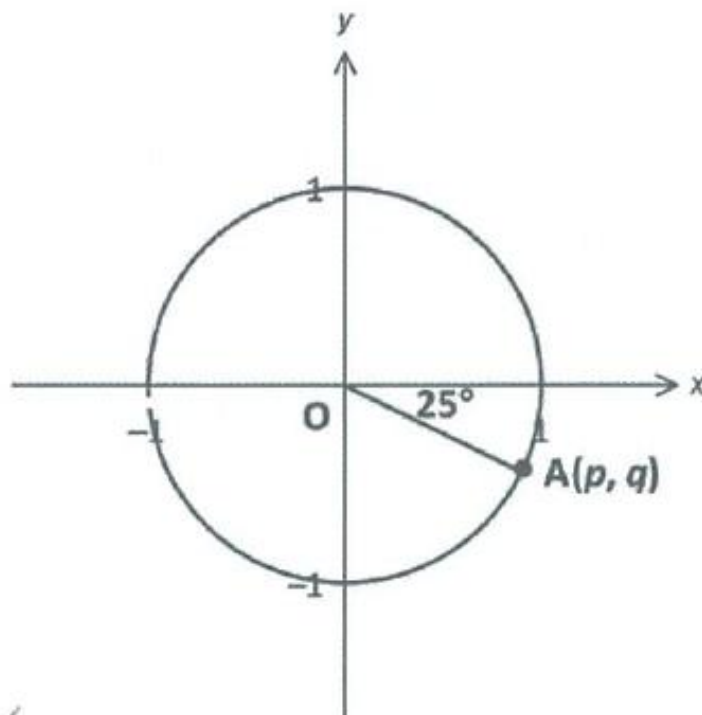
b) $\sin (25^\circ)$

c) $\cos (155^\circ)$

d) $\sin (205^\circ)$

e) $\tan (115^\circ)$

f) $\tan (-155^\circ)$



Question 5 (1.2.8)**(2, 2, 2 = 6 marks)**

What are the exact values of

a) $\sin\left(-\frac{2\pi}{3}\right)$

b) $\tan\left(\frac{15\pi}{6}\right)$

c) $\cos 210^\circ$

END OF TEST